Transitivity of tableaux derivability

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof cites the existence of closed tableaux from the hypotheses and displays none itself, so no tableau environment occurs in the counted span

Editorial note. The label prop:transitivity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed-formula reading of tableaux derivability. Gamma proves B iff for some finite list C_1,...,C_n from Gamma the signed assumption set {False B, True C_1, ..., True C_n} has a closed tableau.. Both hypotheses are unpacked into closed-tableau existence for finite signed assumption sets.
- The Cut rule. A tableau branch may be split on any sentence A into a branch with True A and one with False A.. The composition step: Cut on A yields two branches, each carrying exactly the assumptions of one of the two given closed tableaux.

Proposition 0.1 (Transitivity)**.** *If* $\Gamma \vdash A$ *and* $\{A\} \cup \Delta \vdash B$*, then* $\Gamma \cup \Delta \vdash B$*.*

Proof. If $\{A\} \cup \Delta \vdash B$, then there is a finite subset $\Delta_0 = \{C_1, \ldots, C_n\} \subseteq \Delta$ such that

$$\{\bot(B), \top(A), \top(C_1), \ldots, \top(C_n)\}$$

has a closed tableau. If $\Gamma \vdash A$ then there are D_1, ..., $D_m \subseteq \Gamma$ such that

$$\{\bot(A), \top(D_1), \ldots, \top(D_m)\}$$

has a closed tableau.

Now consider the tableau with assumptions

$$\bot(B), \top(C_1), \ldots, \top(C_n), \top(D_1), \ldots, \top(D_m).$$

Apply the Cut rule on A. This generates two branches, one has $\top(A)$ in it, the other $\bot(A)$. Thus, on the one branch, all of

$$\{\bot(B), \top(A), \top(C_1), \ldots, \top(C_n)\}$$

are available. Since there is a closed tableau for these assumptions, we can attach it to that branch; every branch through $\top(A)$ closes. On the other branch, all of

$$\{\bot(A), \top(D_1), \ldots, \top(D_m)\}$$

are available, so we can also complete the other side to obtain a closed tableau. This shows $\Gamma \cup \Delta \vdash B$. □